

Indexing and Slicing

```
In [ ]: Indexing # means picking one item as per its position
```

```
In [7]: s = 'Welcome'  
s
```

```
Out[7]: 'Welcome'
```

```
In [8]: s[4]
```

```
Out[8]: 'o'
```

```
In [9]: s[6]
```

```
Out[9]: 'e'
```

```
In [11]: s = 'Welcome'  
s
```

```
Out[11]: 'Welcome'
```

Forward indexing

```
In [12]: s = 'Welcome'  
s
```

```
Out[12]: 'Welcome'
```

```
In [20]: s = 'Welcome'  
s
```

```
Out[20]: 'Welcome'
```

```
In [21]: s[:]
```

```
Out[21]: 'Welcome'
```

Forward Slicing

```
In [22]: s[2:6]
```

```
Out[22]: 'lcom'
```

```
In [23]: s[4:8]
```

```
Out[23]: 'ome'
```

```
In [24]: s[1:5]
```

```
Out[24]: 'elco'
```

Backword Slicing

```
In [26]: s = 'Phone'  
s
```

```
Out[26]: 'Phone'
```

```
In [27]: s[-2]
```

```
Out[27]: 'n'
```

```
In [32]: s[-3:]
```

```
Out[32]: 'one'
```

```
In [31]: s[-1]
```

```
Out[31]: 'e'
```

```
In [34]: s[-2]
```

```
Out[34]: 'n'
```

```
In [35]: s[-2:]
```

```
Out[35]: 'ne'
```

Advance Slicing : Means in python to extract or modify the part of given list using start, step and stop

Start means where to begin

step means how to move (Skipping or jumping)

stop means where to stop (there is no stop included)

```
In [36]: s = 'A','B','C','D','E','F'  
s
```

```
Out[36]: ('A', 'B', 'C', 'D', 'E', 'F')
```

```
In [39]: s[1:6:4]
```

```
Out[39]: ('B', 'F')
```

```
In [48]: s[2:5:2]
```

```
Out[48]: ('C', 'E')
```

